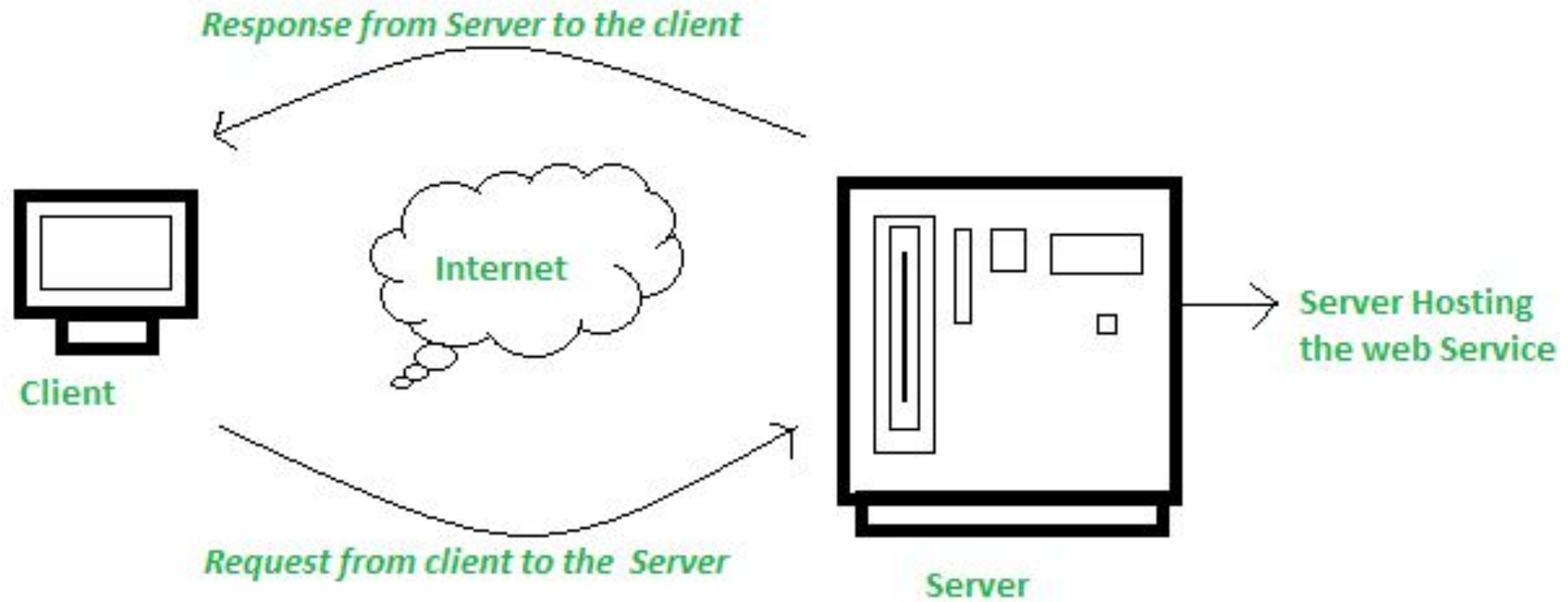


API and Web Services

web service (WS)

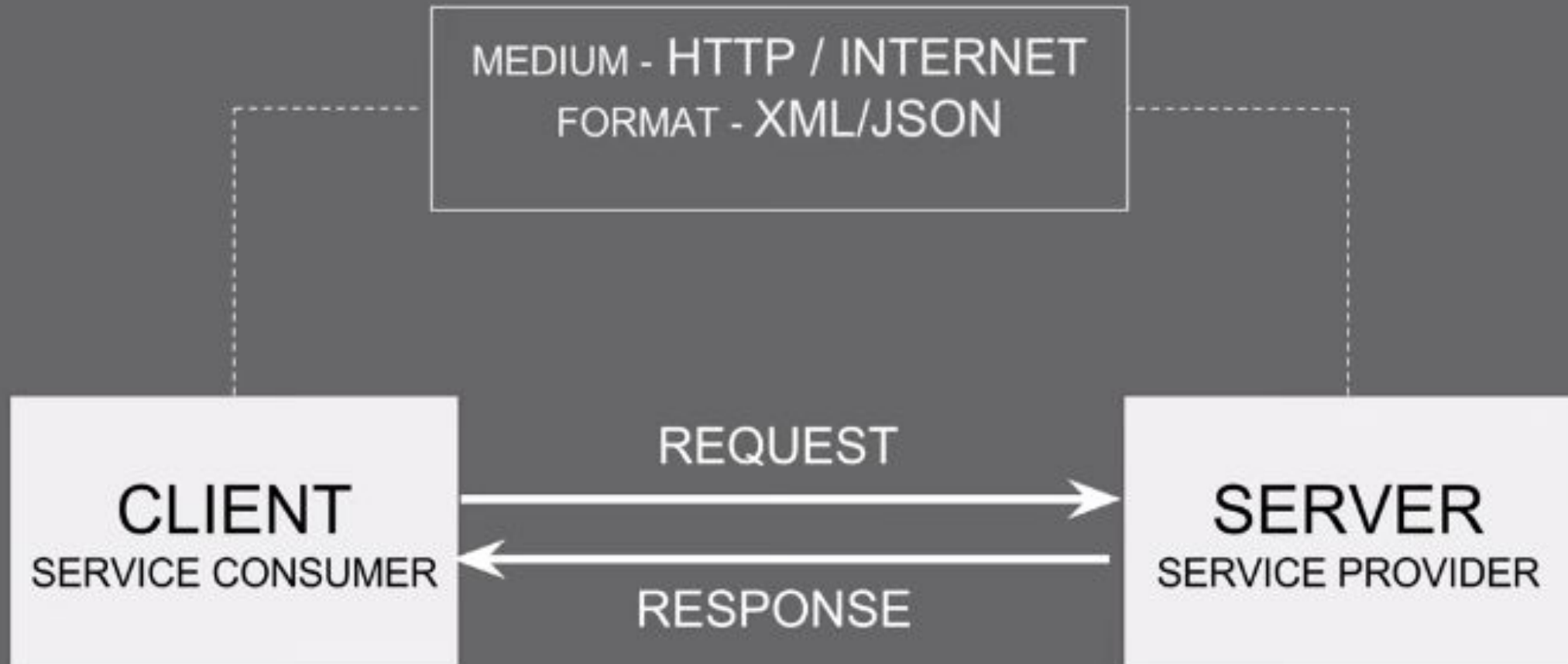
- a service offered by an electronic device to another electronic device, communicating with each other via the Internet, or
- a server running on a computer device, listening for requests at a particular port over a network, serving web documents (HTML, JSON, XML, images).

web service (WS)



WEB SERVICES

Services available over the web



WEB SERVICES

Services available over the web

SOAP

Simple Object Access Protocol

Medium : HTTP (POST)

Format : XML

REST

REpresentational State Transfer

Medium : HTTP (POST, GET, PUT, DELETE, ...)

Format : XML/JSON/TEXT...



Types of web services

- **Extensible Markup Language.** XML is used to tag, code and decode data. XML-remote procedure call, or XML-RPC, is a basic XML protocol using HTTP to exchange data and communicate between client and server systems.
- **Simple Object Access Protocol.** SOAP is an XML-based web service protocol used to transfer data using SOAP messages. The SOAP protocol was developed to enable different programming languages to communicate quickly and with minimal effort using HTTP or Simple Mail Transfer Protocol (SMTP).

- **Web Services Description Language.** WSDL is used to tell the client application what's included in the web service and how to connect. Variations on WSDL include Web Services Conversation Language and Web Services Flow Language.
- **Representational State Transfer.** Based on HTTP, REST provides interoperability between devices and the internet for application programming interface (API)-based tasks. While not all RESTful web services use the REST protocol, applications built with RESTful APIs are more lightweight, manageable and scalable.

- **Universal Description, Discovery and Integration.** UDDI is an XML-based standard that lists and details what services are available in an application. It makes web services discoverable to other services and facilitates digital transactions and e-commerce.

WSDL

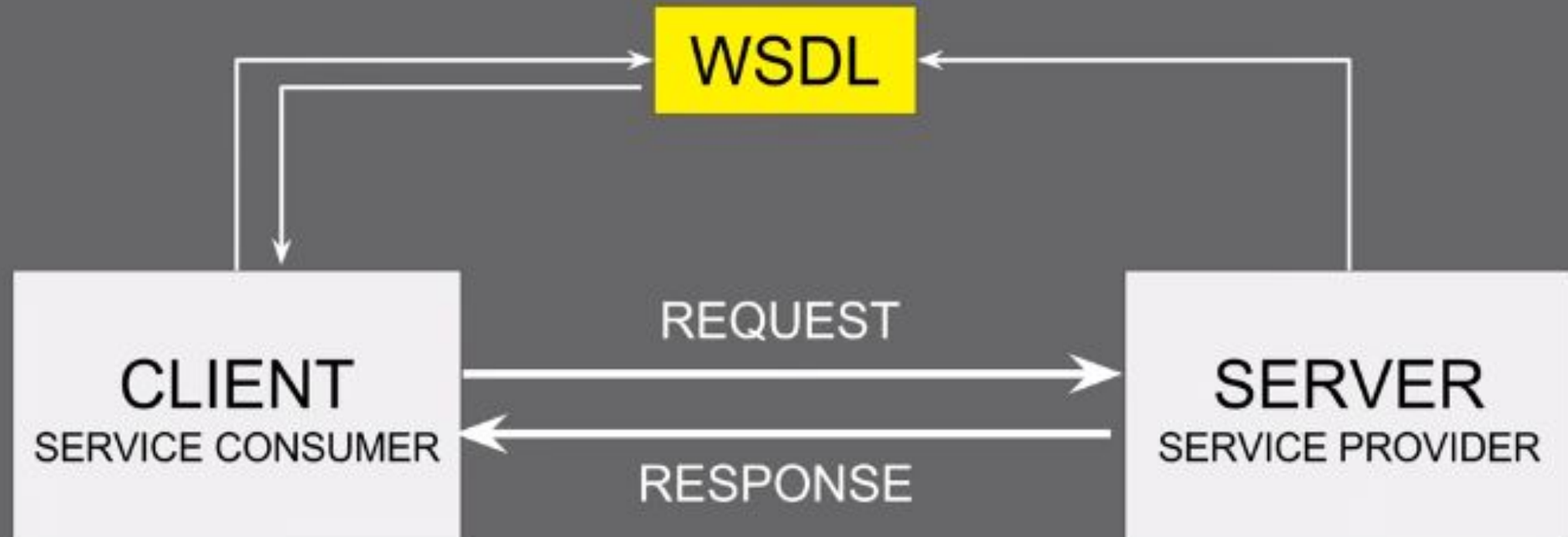
- Service Provider publishes an interface for his web services that describes all attributes of the web services.
- This is XML based interface and is called Web Services Description Language WSDL.
- Describes the functionalities of the web services.

WEB SERVICES

Services available over the web

WSDL

UDDI



WEB SERVICES

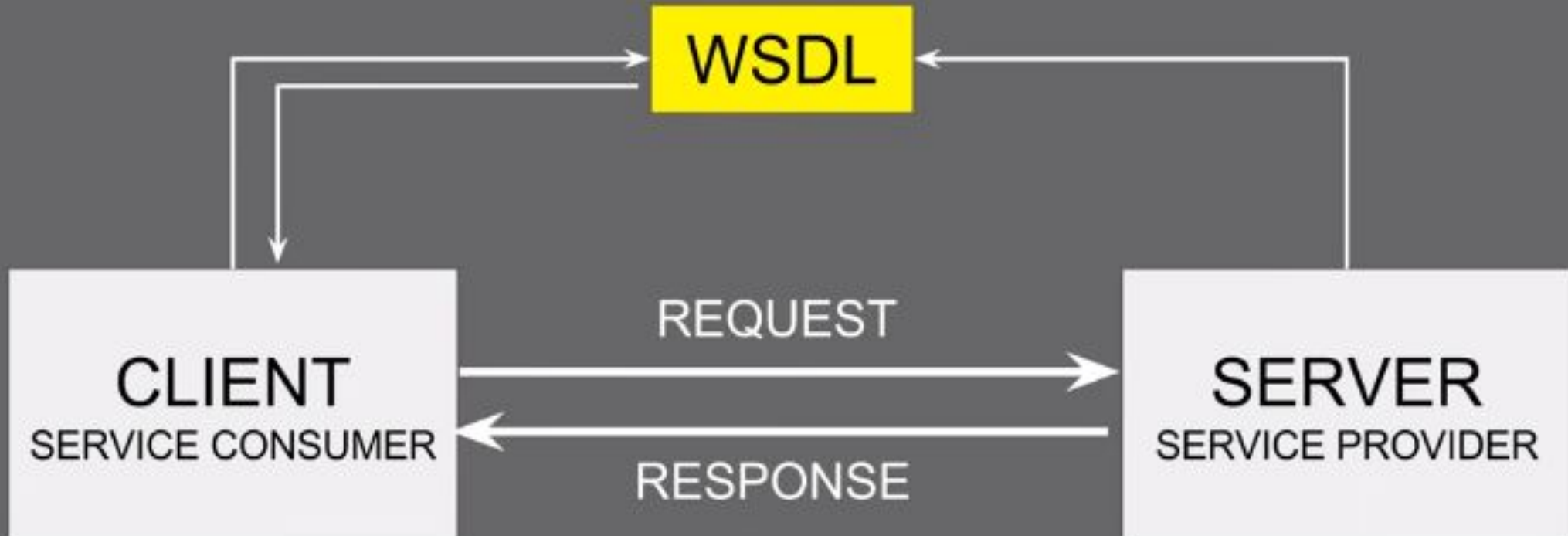
Services available over the web

WSDL

UDDI



WSDL



What are SOAP Web Services ?

A web service that complies to the **SOAP Web Services Specifications** is a SOAP Web Service.

What are these specifications / standards ?

Who defines and dictates these standards ?



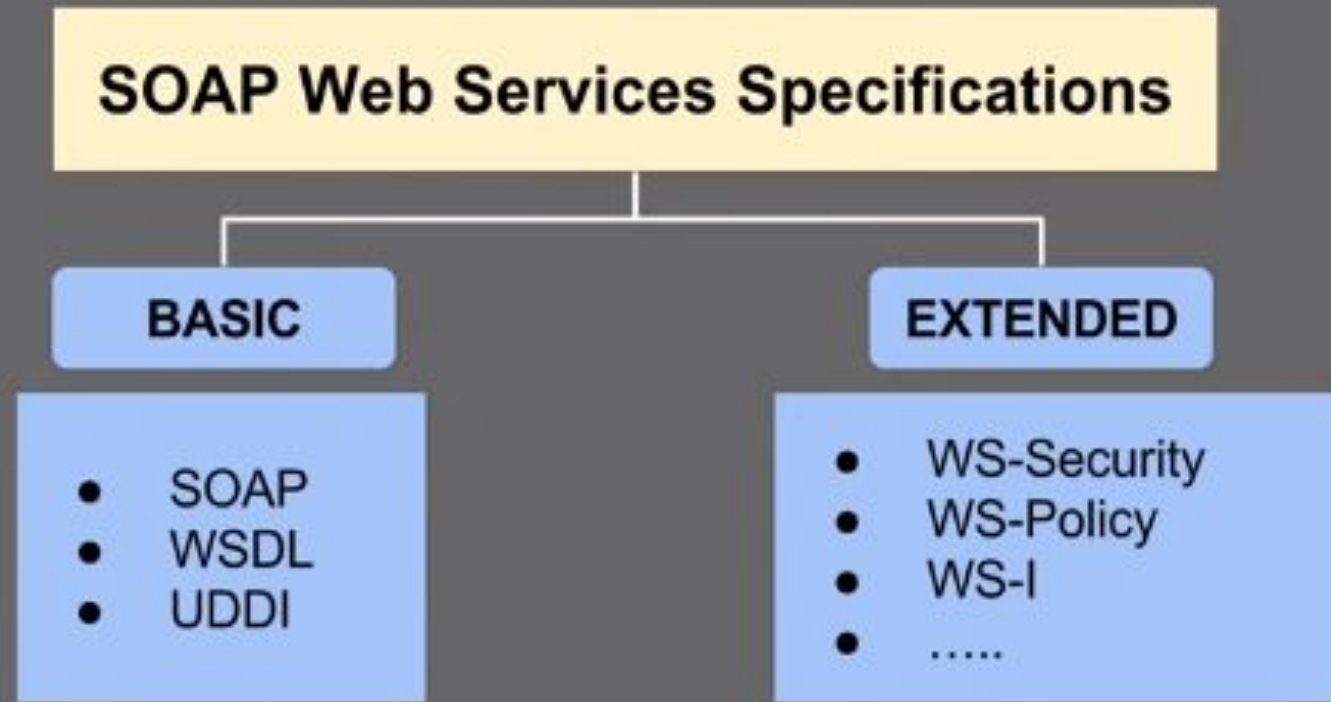
W3C

(World Wide Web Consortium)

An international community that develops open standards for world wide web.

<https://www.w3.org/>

What are SOAP Web Services ?



What are SOAP Web Services ?

SOAP

Simple **O**bject **A**ccess **P**rotocol

(with SOAP v1.2 - the expansion is discontinued)

Protocol / Rules / Definitions how 2 Applications will talk to each other over the web

What are SOAP Web Services ?

SOAP



What are SOAP Web Services ?

SOAP

All information/message exchange happens over a common format:
XML



What are SOAP Web Services ?

SOAP

All information/message exchange happens over a common format:
XML

XML messages have a defined structure:
SOAP MESSAGE



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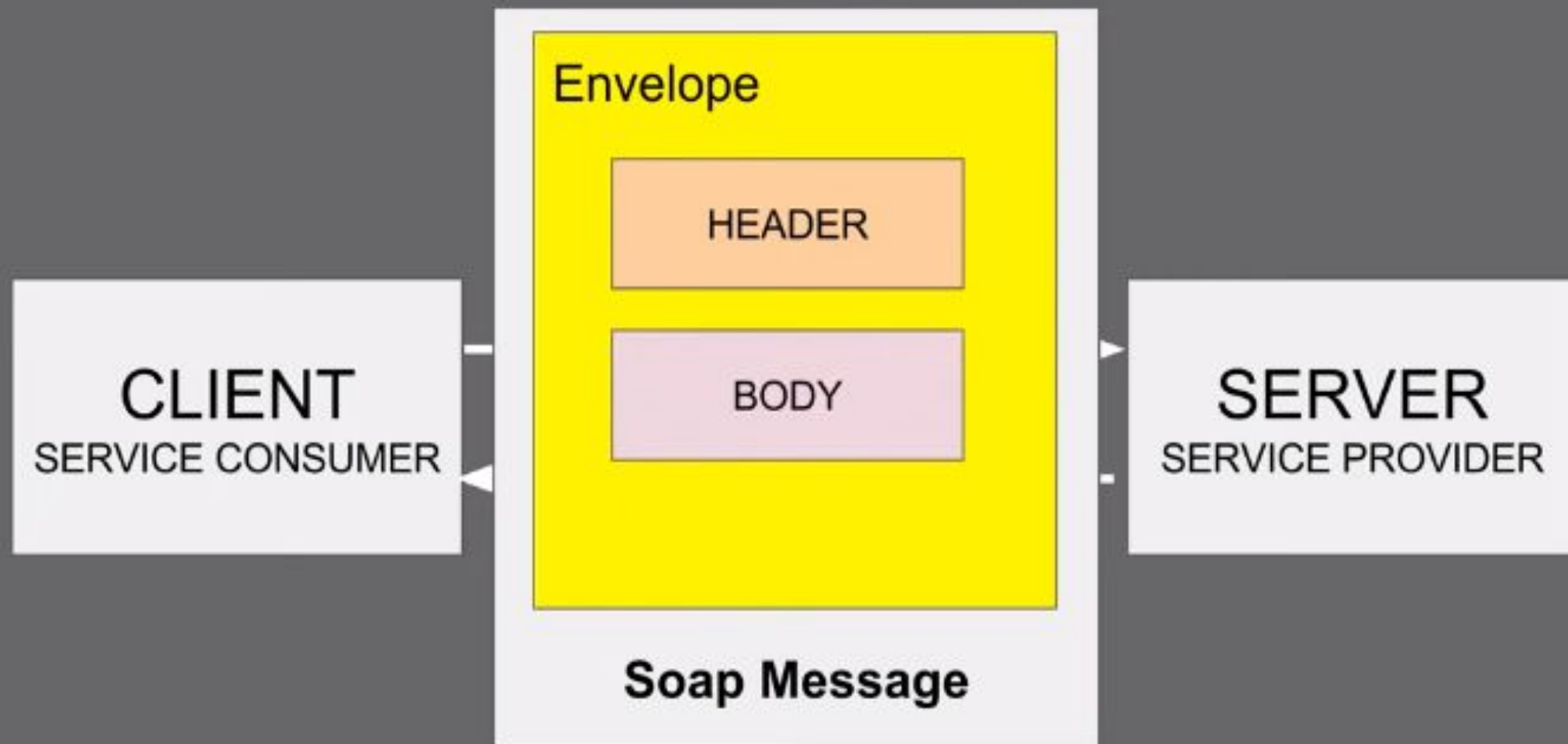
CLIENT
SERVICE CONSUMER

SOAP MESSAGE consist of:
Envelope
Header
Body

SERVER
SERVICE PROVIDER

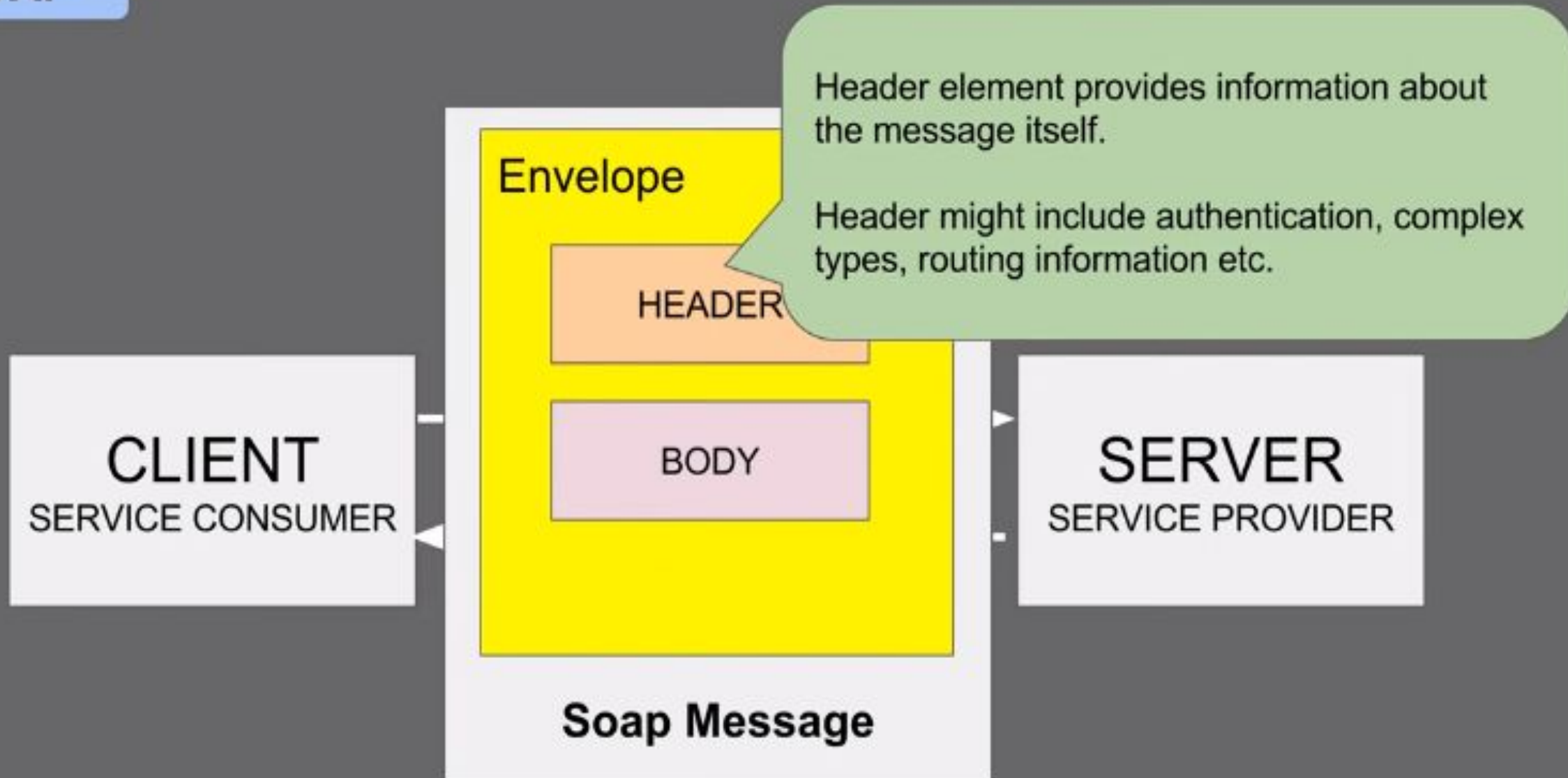
What are SOAP Web Services ?

SOAP



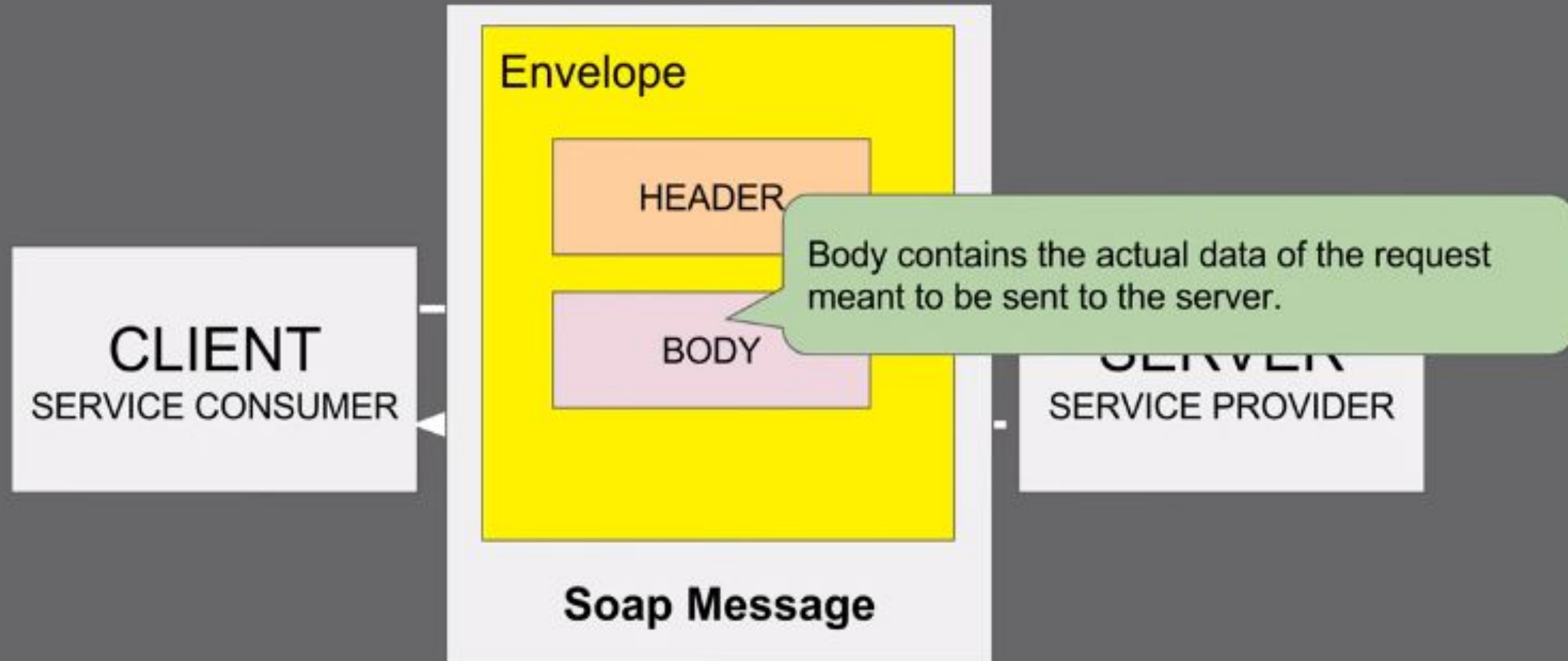
What are SOAP Web Services ?

SOAP



What are SOAP Web Services ?

SOAP



What are SOAP Web Services ?

SOAP

CLIENT
SERVICE

SERVER
PROVIDER

```
<?xml version="1.0" encoding="UTF-8"?>
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/"
  xmlns:c="http://www.acmeOrders.com/OrderService"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance">
  <soap:Body>
    <c:OrderMessage>
      <localElement>
        <FirstName>John</FirstName>
        <LastName>Smith</LastName>
        <Street>High Street</Street>
        <City>London</City>
        <ZipCode>W1A1AA</ZipCode>
        <PartNumber>ABC1234</PartNumber>
        <Quantity>1</Quantity>
      </localElement>
    </c:OrderMessage>
  </soap:Body>
</soap:Envelope>
```

What are SOAP Web Services ?

A web service that complies to the **SOAP Web Services Specifications** is a SOAP Web Service.

SOAP Web Services

- SOAP
- WSDL
- UDDI

All information/message exchange happens over a common format:
XML

XML messages have a defined structure:
SOAP MESSAGE

SOAP MESSAGE consist of:
Envelope
Header
Body

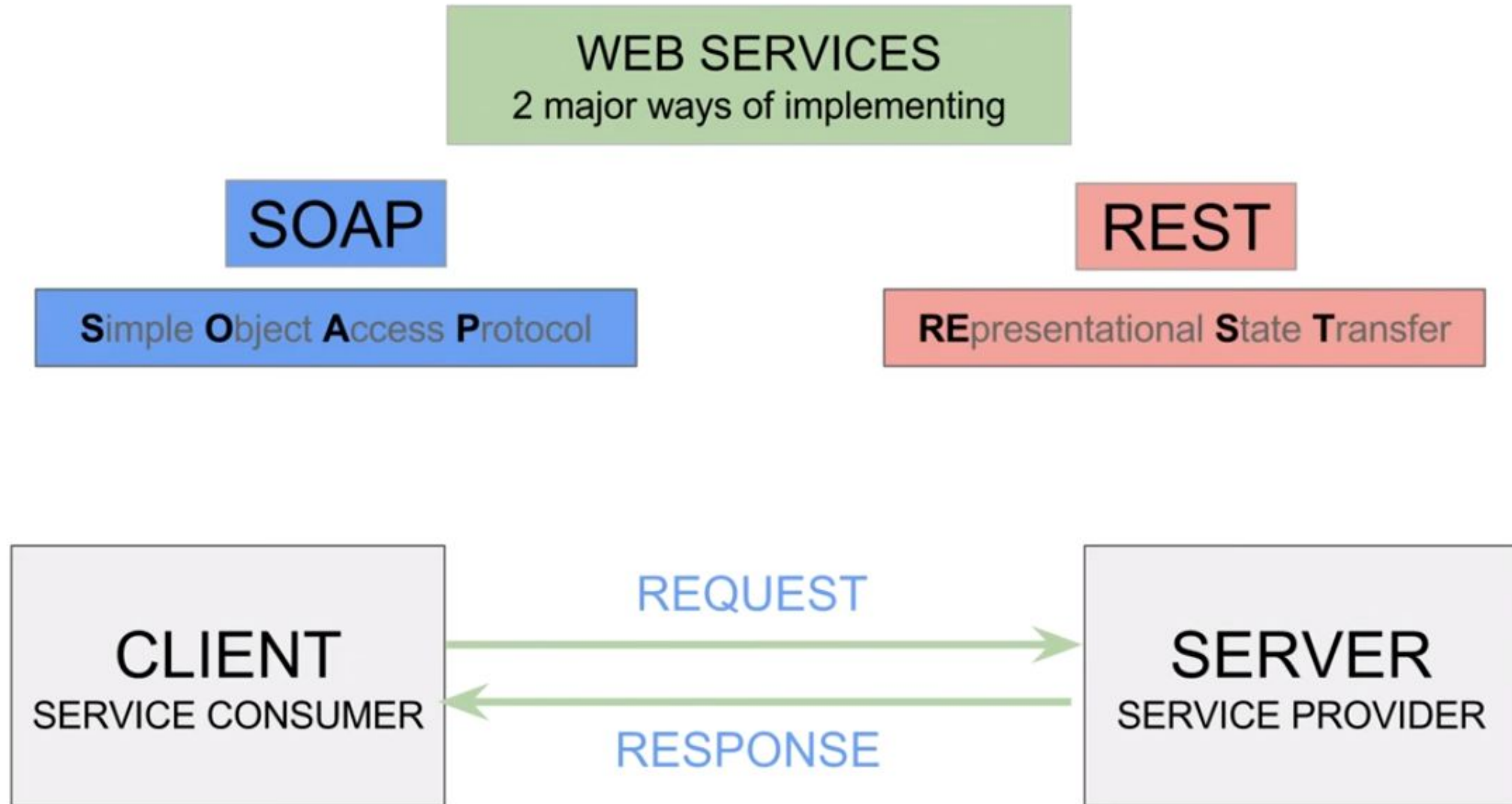
Envelope

HEADER

BODY

Soap Message

What are RESTful Web Services ?



What are RESTful Web Services ?

A web service that communicates / exchanges information between 2 applications using **REST architecture/principles** is called a **RESTful Web Service**.

What is REST ?

What rules should a web service follow to become a RESTful Web Service ?

What are RESTful Web Services ?

What is REST

REST (REpresentational State Transfer)

Is an architectural style.

REST defines a set of principles to be followed while designing a service for communication / data exchange between 2 applications.

When these principles are applied while designing web services (for client - server interactions) we get :

RESTful Web Services.



What are RESTful Web Services ?

What are the principles/constraints of REST architecture

Uniform Interface

Stateless

Cacheable

Layered System

Code On Demand

(Optional)

What are RESTful Web Services ?

What are the principles of RESTful Web Services ?

Uniform Interface

1. **Resource** : everything is a resource
2. **URI** : any resource/data can be accessed by a URI
3. **HTTP** : make explicit use of HTTP methods

Stateless

4. **Stateless** : all client-server communications are stateless

Cacheable

5. **Caching** : happens at client side

Layered System

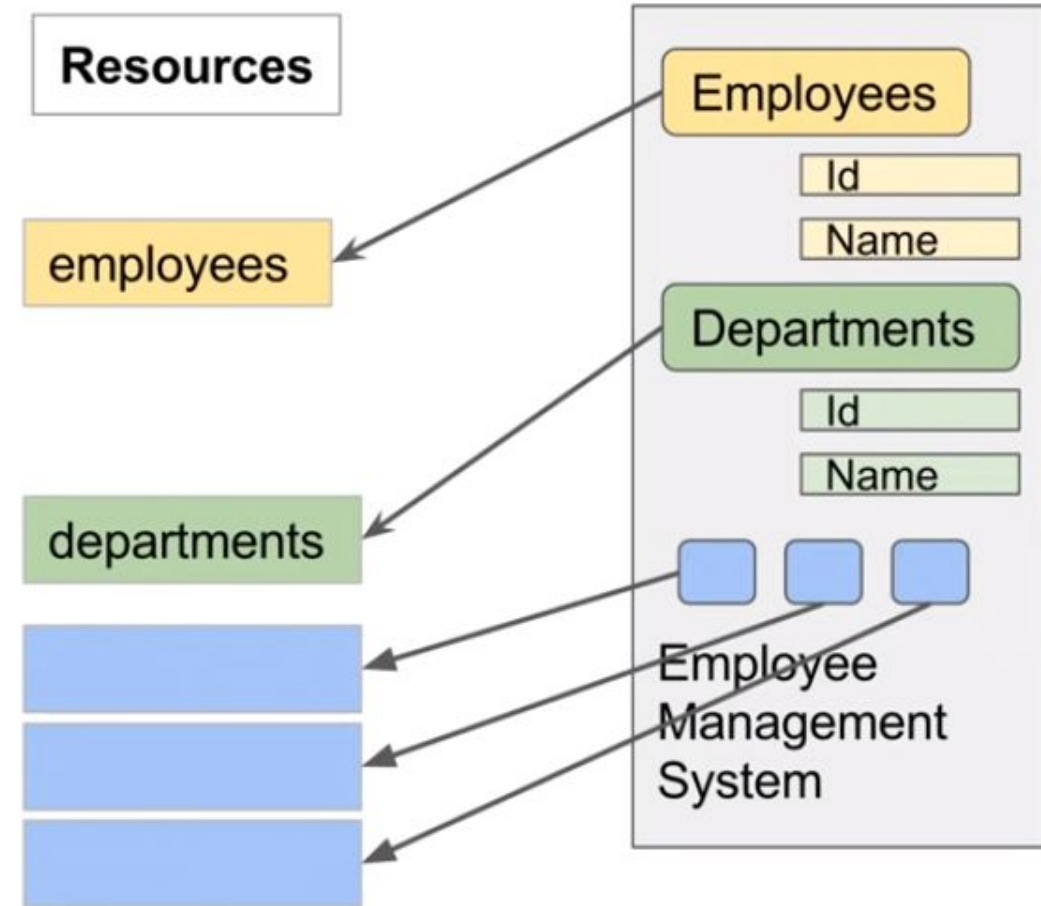
6. **Layering** : layers can exist between client and server

Code On Demand
(Optional)

7. **Code-On-Demand** : ability to download and execute code on client side.

What are RESTful Web Services ?

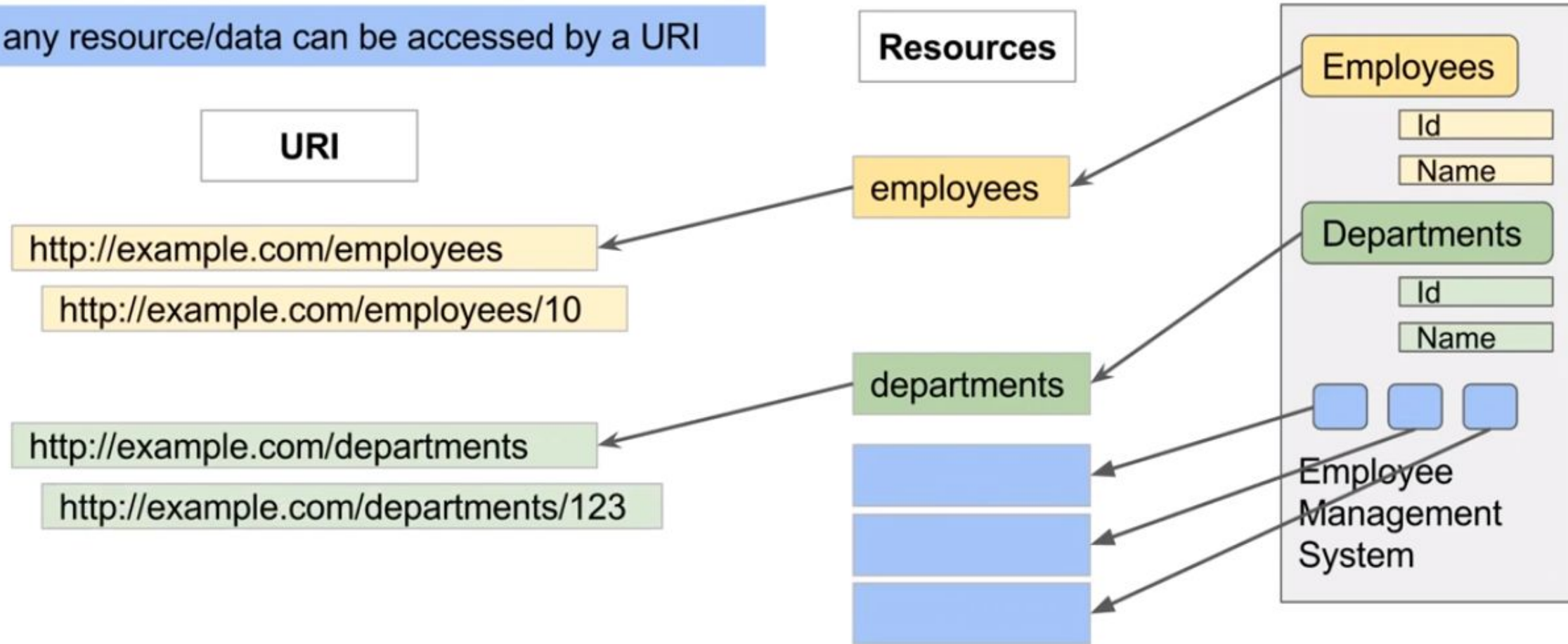
1. Resource : everything is a resource



What are RESTful Web Services ?

1. **Resource** : everything is a resource

2. **URI** : any resource/data can be accessed by a URI



What are

How to get list of employees
from a particular department
?

1. **Resource** : everything is a resource

2. **URI** : any resource/data can be accessed by

URI

<http://example.com/employees>

<http://example.com/employees/10>

<http://example.com/departments>

<http://example.com/departments/123>

<http://example.com/departments/123/employees>

employees

departments

Employees

Id

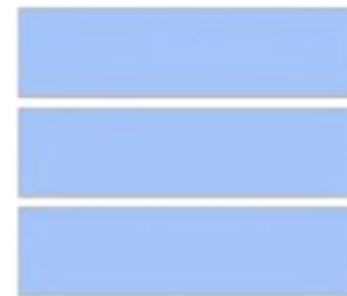
Name

Departments

Id

Name

Employee
Management
System



What are RESTful Web Services ?

1. Resource :

2. URI : any re

“The key abstraction of information in REST is a resource. Any information that can be named can be a resource: a document or image and so on..” - Roy Fielding

http

ht

<http://example.com/departments>

<http://example.com/departments/123>

<http://example.com/departments/123/employees>

Employees

Id

Name

Departments

Id

Name

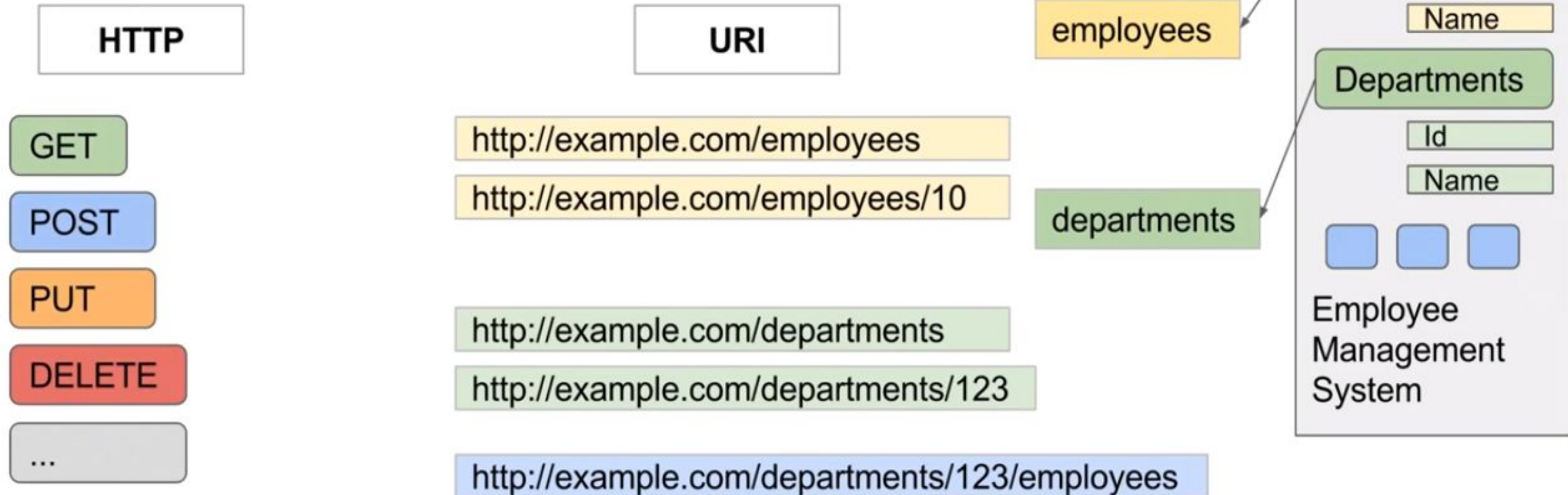
Employee
Management
System

What are RESTful Web Services ?

1. **Resource** : everything is a resource

2. **URI** : any resource/data can be accessed by a URI

3. **HTTP** : make explicit use of HTTP methods

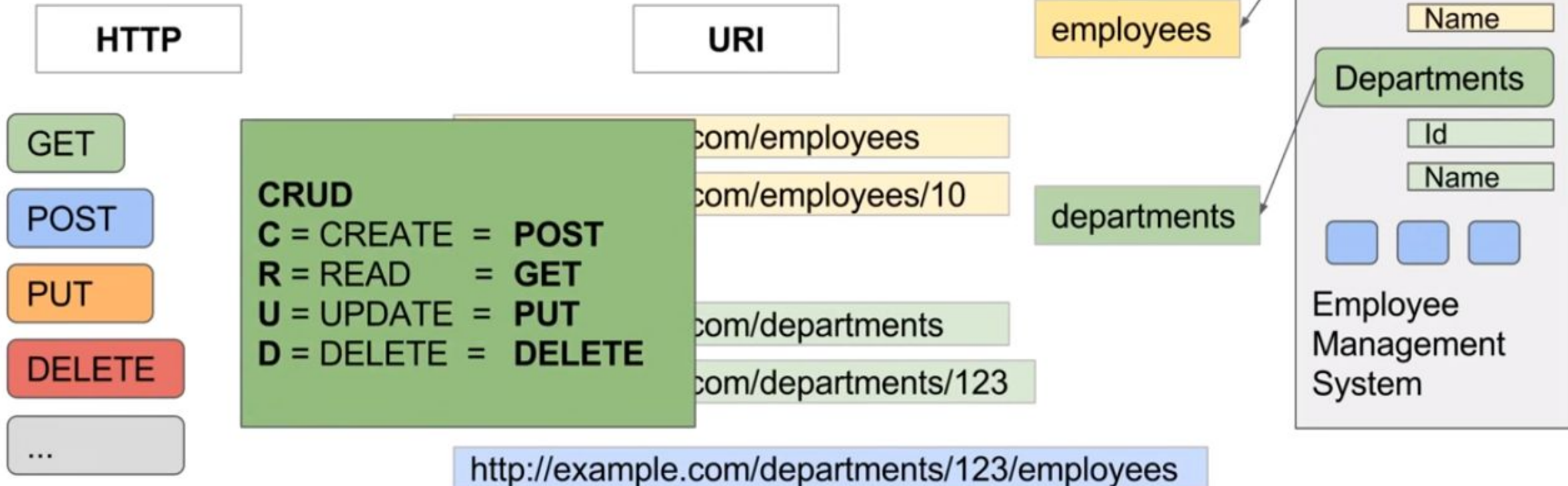


What are RESTful Web Services ?

1. **Resource** : everything is a resource

2. **URI** : any resource/data can be accessed by a URI

3. **HTTP** : make explicit use of HTTP methods



What are RESTful Web Services ?

1. Resource

2. URI

3. HTTP

Using HTTP Methods along with URI, we can access/modify any resource or resource information.

REQUEST

GET - <http://example.com/employees>
GET - <http://example.com/employees/10>
DELETE - <http://example.com/employees/10>

POST - <http://example.com/employees>
+
Data of new employee

PUT - <http://example.com/employees/10>
+
Data to be changed

RESPONSE

list of employees
details of employee with id=10
deletes employee with id=10

id of new employee

modifies data for employee 10

employees

id

Name

employees

id

Name

employee

employee

What are RESTful Web Services ?

1. Resource : everything is a resource

2. UR

3. HT

GET - <http://example.com/employees/10> details of employee with id=10

HTTP

GET

POST

PUT

DELETE

Verbs

RESOURCE

employees

departments

Nouns

<http://example.com/getemployees/10> x

[http:// GET example.com/employees/10](http://example.com/employees/10) ✓

What are RESTful Web Services ?

1. Resource **UNIFORM INTERFACE - 1st principle of REST**

2. URI : **Identification of Resources - Resources + URI + HTTP**

Each resource has a URI and is accessed through a defined set of HTTP methods (GET, PUT, POST, DELETE)

Manipulation of Resources using Representations

Each resource can have one or more representations. Such as application/xml, application/json, text/html, etc. Clients and servers negotiate to select representation.

Self descriptive messages

Requests and responses contain not only data but additional headers describing how the content should be handled. Such as if it should be cached, authentication requirements, etc.

`http://example.org/departments/123`

`http://example.org/departments/123/employees`

Employees

Id

Name

Departments

Id

Name

Employee
Management

System

What are RESTful Web Services ?

SOAP

```
<?xml version="1.0"?>
<soap:Envelope
  xmlns:soap="http://www.w3.org/2001/12/soap-envelope"
  soap:encodingStyle="http://www.w3.org/2001/12/soap-encoding">
  <soap:body pb="http://www.example.com/employees">
    <pb:GetEmployee>
      <pb:EmpId>123</pb:EmpId>
    </pb:GetEmployee>
  </soap:Body>
</soap:Envelope>
```

REST

<http://example.com/employees/123>

GET

What are RESTful Web Services ?

4. **Stateless** : all client-server communications are stateless

Each request from the client to the server must contain all of the data that is necessary to handle the request.
No need of storing any state on the server.



What are RESTful Web Services ?

4. Stateless

STATELESSNESS

Server should not require to store the state of a session.

If the concept of a session is required all information should be stored at client side and sent with every subsequent request.

This makes every request (call) from client independent and a complete entity in itself.

For Example:

If a person is adding items to his cart on a shopping website, the state of the cart should be stored at the client side only.

And this state of the cart has to be sent from the client to the server whenever required.



What are RESTful Web Services ?

4. **Stateless** : all client-server communications are stateless

Each request from the client to the server must contain all of the data that is necessary to handle the request.
No need of storing any state on the server.

+

Improves Web Service Performance



What are RESTful Web Services ?

5. Caching : happens at client side

The data within a response to a request must be implicitly or explicitly labeled as cacheable or non-cacheable



What are RESTful W

5. Caching : happens at client side

The data within a response to a request must be implicitly labeled as cacheable or non-cacheable

Server generates responses that indicate whether they are cacheable or not to improve the performance by reducing number of requests for duplicate resources.

Server does this by including a **Cache-Control** and **Last-Modified** (date value) In HTTP Response Headers



What are RESTful W

5. Caching: happens at client side

Server generates responses that indicate

URL	Status	Domain	Size
▼ GET 2	304 Not Modified	127.0.0.1:9292	1.4 KB

Headers Response Cache HTML Cookies

Response Headers

[view source](#)

Age 100
Cache-Control max-age=180, public
Connection close
Date Wed, 26 Sep 2012 09:41:16 GMT
Etag "2fd5257a3dd12a2b0d130931f3bc1ab5"
Server thin 1.4.1 codename Chromeo
X-Content-Digest 7ce9d1010d836cf77d921adc615ed387d5a59c91
X-Rack-Cache fresh
X-Request-Id ca28f6820243fe9fac8c7516a01c5edd
X-Runtime 0.122089
x-ua-compatible IE=Edge

ources.

5

Client uses the **Cache-Control** header to determine whether to cache the resource (make local copy) or not.

RESTful W

implicitly

Server generates responses that indicate whether they are cacheable or not to improve the performance by reducing number of requests for duplicate resources.

Server does this by including a **Cache-Control** and **Last-Modified** (date value) in HTTP Response Headers

In RESTful Web Services, This co-ordination between **Client** and **Server** is essential for being **Stateless**.

CLIENT
SERVICE CONSUMER

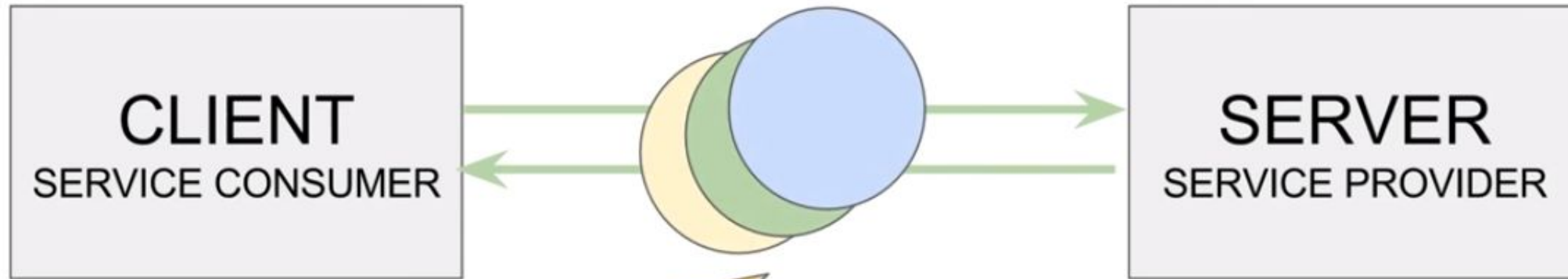
REQUEST

RESPONSE

SERVER
SERVICE PROVIDER

What are RESTful Web Services ?

6. Layering : multiple layers can exist between client and server



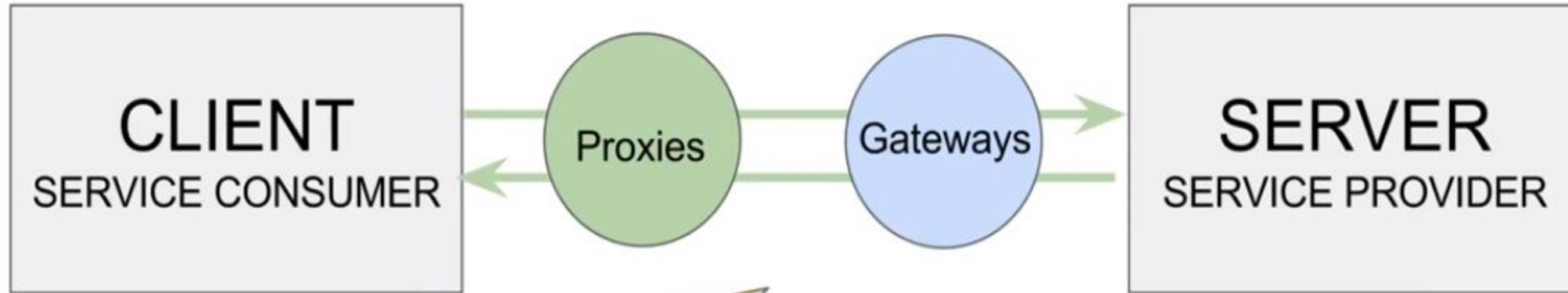
There can be many layers (intermediaries) between client and server.
These are HTTP Intermediaries.

Can be used for message translations / improving performance with caching etc.

Can include Proxies and Gateways

What are RESTful Web Services ?

6. Layering : multiple layers can exist between client and server



Client and Server communication channel may also include intermediaries like Proxies and Gateways.

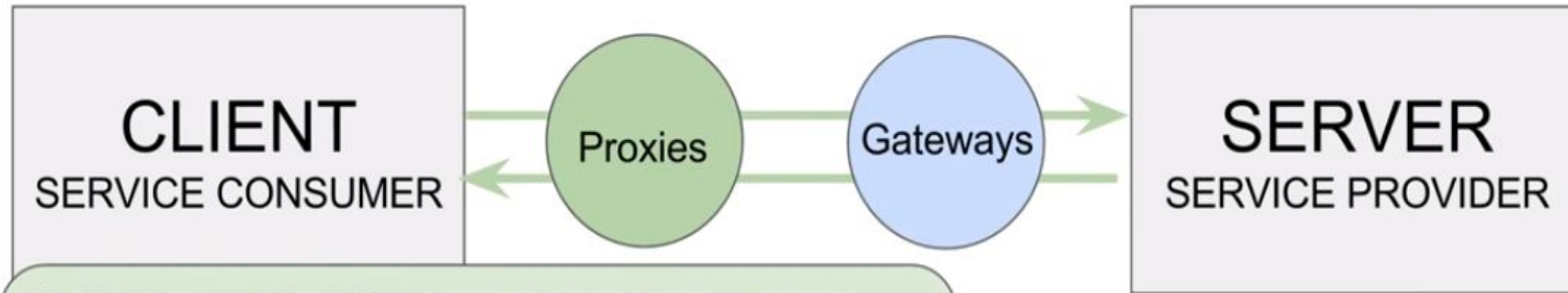
Proxies - are chosen by client

Gateways - are chosen by server

There can be several proxies and gateways.

What are RESTful Web Services ?

6. Layering : multiple layers can exist between client and server



A client may send the request to a Proxy server (instead of main server).
Proxy server evaluates the request to simplify or control its complexity, etc.

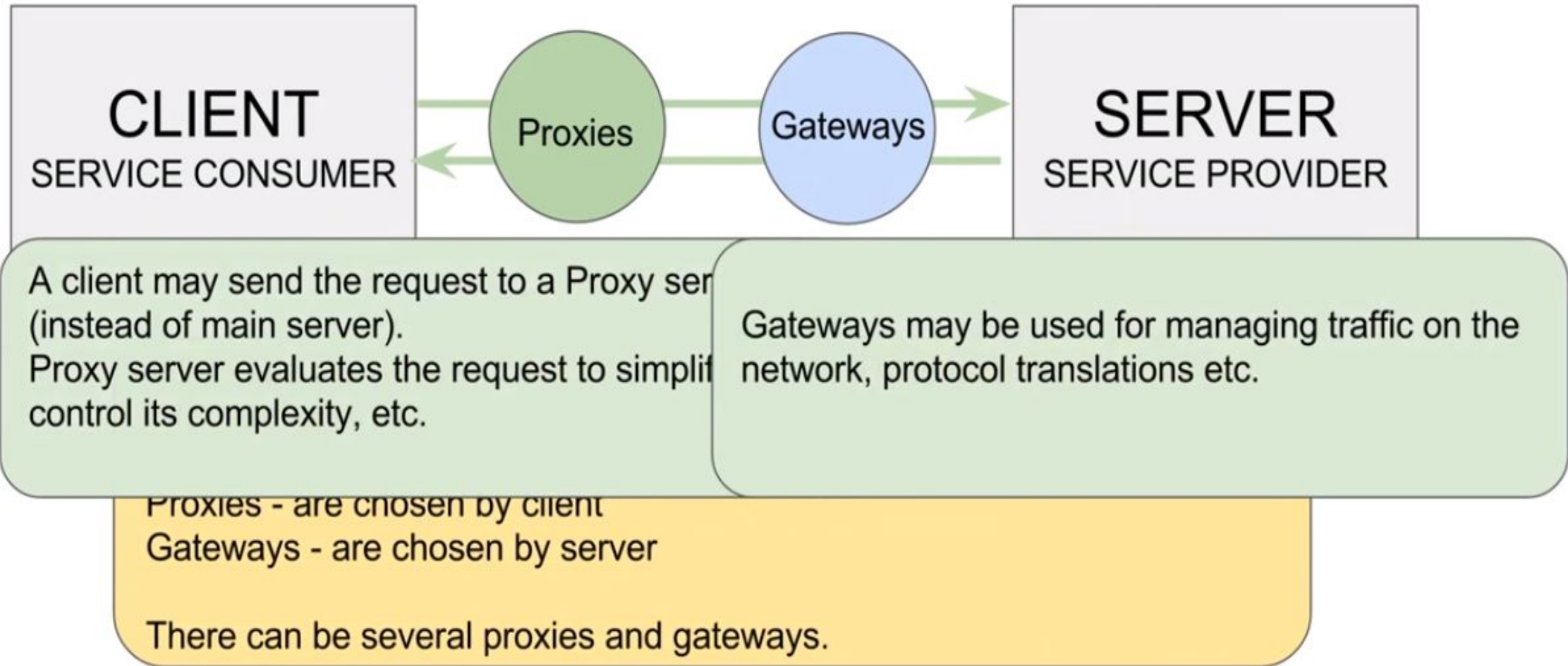
also include

Proxies - are chosen by client
Gateways - are chosen by server

There can be several proxies and gateways.

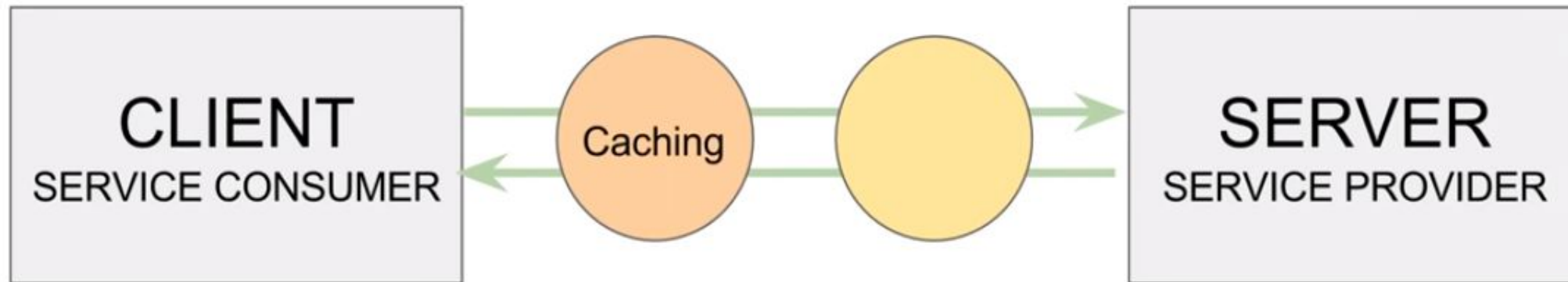
What are RESTful Web Services ?

6. Layering : multiple layers can exist between client and server



What are RESTful Web Services ?

6. Layering : multiple layers can exist between client and server



Example : we use a layer (intermediary) that can cache a response and store it for an hour.

If a new request comes from the client within an hour, the cached response will be sent from the intermediary (without going until the server)

Improves Performance and Scalability.

What are RESTful Web Services ?

6. Layering : multiple layers can exist between client and server

This is to improve scalability.

The layered constraint was added to address improving internet sized scalability requirements. Each layer cannot see beyond the immediate layer with which it is communicating with. This places boundaries on the overall complexity of the system.

Like most solutions for scalability there are usually trade-offs. Latency is increased with the introduction of layers but the cache-constraint above can certainly help reduce the amount of requests over the network.

What are RESTful Web Services ?

7. **Code-on-Demand** : ability to download and execute code at client side

The code-on-demand constraint allows clients to be extendable by downloading and executing code. Similarly to java script in a web browser this allows you to add functionality without re-deploying client software.

The code-on-demand constraint is ***optional***.

What are RESTful Web Services ?

7. **Code-on-Demand** : ability to download and execute code at client side

The code-on-demand constraint allows clients to be extendable by downloading and executing code. Similarly to java script in a web browser this allows you to add functionality without re-deploying client software.

The code-on-demand constraint is ***optional***.

For example:

Client requests a resource
Server returns resource with some JavaScript

What are RESTful Web Services ?

REST REpresentational State Transfer

REPRESENTATION = description of the current state of the resource

transferred between server and client.

GET /images/1
Accept : JPEG

Or GET /images/1.jpg



XML
JSON
HTML
YAML
.....

GET /images/1
Accept : XML

Or GET /images/1.xml



```
<pic>
  <name>mydog</name>
  <size>1080p</size>
</pic>
```

What are RESTful Web Services ?

What is REST

REST (REpresentational State Transfer)

Is an architectural style.

REST defines a set of principles to be followed while designing a service for communication / data exchange between 2 applications.

When these principles are applied while designing web services (for client - server interactions) we get :

RESTful Web Services.

What are RESTful Web Services ?

What is REST

REST (REpresentational State Transfer)

SET OF RULES to guide data exchange / communication between 2 applications.

Principles to guide how the data/information (STATE) of a Resource is transferred using REPRESENTATIONS.